Red ASM Security

Software Test Plan (STP) & Software Test Report (STR)

SP-108-Red-ASM

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	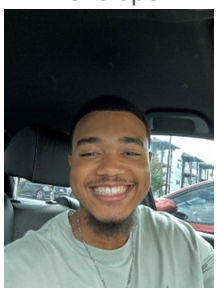	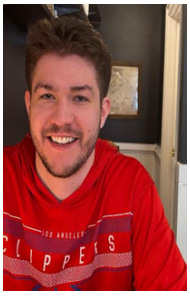	

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1. Overview

1.1. Purpose

The purpose of this document is to describe the testing process for our Red ASM Security project. This includes what parts of the system will be tested, how testing will be performed, and how results are going to be recorded. The goal is to make sure the system works correctly and handles security events as expected.

1.1. Scope

This test plan focuses on the main features of the ASM system. These include alert generation, automated responses, and case management. The document explains how these features will be tested and what is expected from each part of the system.

2. Testing Summary

2.1. Scope of Testing

2.1.1. In Scope

The following features are included in testing:

- Slack alerts and case messages
- Slack action buttons such as Block IP and Contain Path
- Shuffle workflows and automation
- Python scripts used for backend logic
- MySQL database data storage
- Custom Wazuh alerts
- Case promotion logic
- Alert threading in Slack

2.1.2. Out of Scope

The following areas are not included in testing:

- Authentication and login systems
- Full integration setup of external tools
- Denial of service attacks
- Accuracy of external scanning tools such as Nuclei or Subfinder

3. Test Objectives

The goal of testing is to make sure the system behaves correctly during both normal use and attack scenarios

- Ensure the Block IP button successfully blocks a malicious IP address
- Ensure the Contain Path button blocks SQL injection attempts
- Ensure normal user queries are not blocked
- Ensure alerts are promoted to cases when required
- Ensure cases include the correct links and action buttons
- Ensure the ASM scanning script uploads data to the database
- Ensure old alerts are removed after a set period of time
- Ensure critical alerts are automatically to cases
- Ensure SQL injection attacks trigger correct alerts
- Ensure alerts and cases are properly threaded in Slack

4. Test Cases

Test Case ID	Description	Expected Result
TC-01	Click Block IP Button	IP is blocked
TC-02	Perform SQL injection attack	Request is blocked
TC-03	Perform normal search query	Request succeeds
TC-04	Trigger alert threshold	Alert becomes case
TC-05	Trigger critical alert	Auto promoted to case
TC-06	Run ASM scan	Data stored in database
TC-07	Wait and generate new alert	Old alerts removed
TC-08	Perform SQL injection variations	Correct alerts triggered
TC-09	Generate multiple alerts	Alerts are threaded correctly
TC-10	Verify case investigation links and action buttons	Cases include correct links and buttons

5. Test Procedures

5.1. TC-01 Block IP

Step 1. Click the Block IP button in Slack
Step 2. Check the Shuffle workflow runs
Step 3. Verify a response message appears in Slack
Step 4. Run sudo iptables -L INPUT -n

Expected result: The IP is listed as blocked

5.2. TC-02 SQL Injection Block (Contain Path)

Step 1. Perform an SQL injection attack
Step 2. Click the Contain Path button
Step 3. Check the Shuffle workflow runs
Step 4. Try the attack again

Expected result: Request is blocked and returns a 403 page

5.3. TC-03 Normal Query Not Blocked

Step 1. Use /rest/products/search?q=
Step 2. Enter normal searches like apple or banana
Step 3. Submit the request

Expected result: Results are returned normally

5.4. TC-04 Alert Promotion to Case

Step 1. Trigger different alerts
Step 2. Check Slack for alert messages
Step 3. Verify alerts meeting threshold become cases

Expected result: Alerts are promoted to cases correctly

5.5. TC-05 Critical Alert Auto Promotion

Step 1. Trigger a critical alert such as data breach

Step 2. Check Slack output
Step 3. Verify case creation

Expected result: Alert is automatically promoted to a case

5.6. TC-06 ASM Scan Database Upload

Step 1. Run ASM scanning script
Step 2. Check for errors
Step 3. Check database entries

Expected result: Data is stored in the database

5.7. TC-07 Alert Removal

Step 1. Perform an attack from an IP
Step 2. Verify alert appears
Step 3. Wait one hour
Step 4. Perform another attack with same IP

Expected result: After one hour the old alert is removed from the cache and only recent alerts remain

5.8. TC-08 SQL Injection Detection

Step 1. Perform SQL injection using OR 1=1
Step 2. Perform SQL injection using UNION or sqlite_master
Step 3. Check alerts in Slack
Step 4: Verify each attack triggers correct custom Wazuh alerts such as general alerts, schema exposure cases, and critical data breach cases

Expected result: Correct alert type is generated for each attack

5.9. TC-09 Alert Threading

Step 1. Perform multiple attacks close together
Step 2. Perform attacks spaced apart
Step 3. Check Slack threads

Expected result: Similar alerts are grouped into threads

5.10. TC-10 Case Links and Actions

Step 1. Trigger an attack that becomes a case
Step 2. Open the case in Slack
Step 3. Check for investigation links
Step 4. Check for action buttons

Expected result: Cases include the correct investigation links and action buttons based on the attack type

6. Test Environment

6.1. Hardware

The system uses virtual machines for testing. One machine is used as the target system and another is used as the manager system.

6.2. Software

The following software is used during testing

- Wazuh
- Shuffle
- Slack
- Python Scripts
- MySQL database

6.3. Network Setup

All systems are connected over a network so they can communicate with each other. This allows alerts, logs, and responses to be shared between components.

7. Test Data

7.1. Attack Data

- SQL injection inputs such as OR and 1=1

- UNION and SELECT queries
- Sqlite master queries
- Brute force login attempts
- File integrity attacks

7.2. Normal USer Data

- Product search queries such as apple, banana, test
- Normal user activity with no malicious input

8. Test Strategy

8.1. Testing Responsibility

Testing will mainly be handled by the project team. Most testing will be done by the team during development. Security testing will involve both the team and external tools since attacks are being simulated using real techniques.

8.2. Testing Approach

8.2.1. Functional Testing

This testing focuses on making sure all features work correctly. This includes Slack alerts, action buttons like Block IP and Contain Path, and case creation.

8.2.2. Integration Testing

This testing checks that all parts of the system work together. This includes communication between Slack, Shuffle, Python scripts, and the database.

8.2.3. Security Testing

This testing focuses on how the system handles attacks. This includes SQL injection, brute force attempts, and file integrity changes.

8.2.4. System Testing

This testing looks at the full system as a whole. It ensures alerts are detected, processed, and responded to correctly.

8.3. Test Execution Plan

Testing will be done in stages with the individual features such as buttons and alerts will be tested first and then the full system testing will be performed. High priority features such as attack detection and automated responses will be tested first since they are the most important parts of the system.

8.4. Testing Requirements

To complete testing the following access is needed:

- Slack workspace access
- Shuffle platform access
- Access to virtual machines
- Database access
- Ability to perform attack simulations

8.5. Resource and Skills

Team members performing testing should have:

- Basic understanding of cybersecurity concepts
- Knowledge of SQL injection and brute force attacks
- Experience with Python scripts
- Ability to read and understand system output

9. Software Test Report (STR)

Requirement	Pass	Fail	Severity
TC-01 Block IP	Yes		Critical
TC-02 SQL injection	Yes		Critical
TC-03 Search query	Yes		Moderate
TC-04 Trigger alert	Yes		Moderate
TC-05 Trigger critical alert	Yes		Critical
TC-06 Run ASM scan	Yes		Moderate
TC-07 Generate new alert	Yes		Minor

TC-08 SQL injection variations	Yes		Critical
TC-09 Generate multiple alerts	Yes		Minor
TC-10 Case links and actions	Yes		Moderate

10. Assumptions and Dependencies

10.1. Assumptions

It is assumed that all tools are running correctly and are properly connected.

10.2. Dependencies

Testing depends on working connections between system components and access to required tools.

11. Entry and Exit Criteria

11.1. Entry Criteria

Testing will begin once the system is fully set up and all components are connected. Test data must also be created.

11.2. Exit Criteria

Testing will end once all test cases have been completed and results have been recorded. Any major issues should be fixed before completion.

12. Tools Used

The following tools are used during testing:

- Slack
- Shuffle
- Wazuh
- Python
- MySQL